1 What is Node.js and why is it used in backend development?

Node.js is a runtime environment that allows JavaScript to run on the server side. It is used in backend development because it is fast, scalable, and uses a non-blocking, event-driven architecture, which is ideal for building APIs, real-time applications, and web servers.

2 Explain the difference between synchronous and asynchronous functions in Node.js.

Synchronous functions execute line by line and block the execution until the task is completed. Asynchronous functions do not block execution; they run in the background and use callbacks, promises, or async/await to handle results, making applications faster and more efficient.

3 What is the purpose of the fs module? Name two methods used for file operations.

The fs (File System) module is used to work with files in Node.js, such as reading, writing, updating, and deleting files. Two commonly used methods are fs.readFile() and fs.writeFile().

4 What is the difference between relative path and absolute path?

A relative path specifies the file location relative to the current directory, while an absolute path specifies the full path starting from the root directory of the system.

5 What is Express and how is it different from Node's core HTTP module?

Express is a web framework for Node.js that simplifies server creation and routing. The core HTTP module requires more manual coding for routing and middleware, whereas Express provides built-in routing, middleware support, and easier API development.

6 What is routing in Express? How do you create a route?

Routing in Express refers to defining how an application responds to client requests for specific endpoints and HTTP methods. A route is created using methods like app.get(), app.post(), app.put(), and app.delete().

7 What is the difference between req.params, req.query, and req.body?

req.params is used to get route parameters from the URL, req. query is used to get query string values from the URL, and req.body is used to get data sent in the request body, usually in POST or PUT requests.

8 What is the response object (res)? Why is it important?

The response object (res) is used in Express to send responses back to the client. It is important because it helps send data, JSON, status codes, and messages to the browser or API client.

9 What is the use of HTTP status codes? Name three commonly used status codes.

HTTP status codes indicate the result of a client's request. Common status codes include 200 (OK), 201 (Created), and 404 (Not Found).

10 What is MVC architecture? Why do we use it?

MVC stands for Model-View-Controller. It is a design pattern used to separate application logic. The Model handles data, the View handles the UI, and the Controller manages the logic. We use MVC to make code organised, scalable, and maintainable.

11 What is MongoDB? How is it different from SQL databases?

MongoDB is a NoSQL database that stores data in JSON-like documents. Unlike SQL databases that use tables and rows, MongoDB uses collections and documents and is schema-flexible and scalable.

12 What is Mongoose and why do we use it in Node.js?

Mongoose is an ODM (Object Data Modelling) library for MongoDB in Node.js. It is used to define schemas, validate data, and simplify database operations.

13 What is a schema in Mongoose?

A schema in Mongoose defines the structure of the documents in a MongoDB collection, including fields, data types, and validation rules.

14 What is a model in Mongoose?

A model in Mongoose is a compiled version of the schema that provides an interface to interact with the database collection for CRUD operations.

15 Which Mongoose methods are used for Create, Read, Update, and Delete operations?

Create: create(), save()
Read: find(), findById()
Update: findByIdAndUpdate(), updateOne()
Delete: findByIdAndDelete(), deleteOne()